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Theological Foundations of Kepler's Astronomy

By Peter Barker and Bernard R. Goldstein***

I. INTRODUCTION

Johannes Kepler (1571–1630) is celebrated today as one of the first defenders of Copernicanism and as the discoverer of three laws of planetary motion. These are usually presented as follows:

First Law: The orbits of the planets are ellipses, with the sun at one focus.

Second Law: The radius vector from the sun to a planet sweeps out equal areas in equal times.

Third Law: The square of a planet's period divided by the cube of its mean distance from the sun is a constant.

Locating these laws in the historical record is surprisingly difficult. The modern reader encounters two sorts of puzzles. First, the clearest statements of each law by Kepler are scattered and in the wrong order; and, considering their present importance, they lack the prominence we would expect them to be given. Second, much of Kepler's work, especially his first defense of Copernicanism as well as the setting of the third law, appears unconnected with modern science. Indeed, much of Kepler's work has been dismissed as mysticism or Neoplatonism. Kepler's frequent and direct statements about religion are also dismissed as, at best, psychologically significant for understanding his scientific achievements.¹

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¹ See (in order of original publication): J. L. E. Dreyer, *History of Planetary Systems from Thales to Kepler* (Cambridge: Cambridge Univ. Press, 1906); Max Caspar, *Kepler*, trans. C. Doris Hellman (1948; New York: Dover, 1993); Herbert Butterfield, *The Origins of Modern Science, 1300–1800* (London: Bell & Sons, 1949); E. J. Dijksterhuis, *The Mechanization of the World Picture*, trans. C. Dikshoorn (1950; Oxford: Clarendon, 1961); Thomas S. Kuhn, *The Copernican Revolution: Planetary Astronomy in the History of Western Thought* (Cambridge, Mass.: Harvard Univ. Press, 1957); Arthur Koestler, *The Sleepwalkers* (1959; New York: Macmillan, 1968); Alexandre Koyré, *The Astronomical Revolution: Copernicus–Kepler–Borelli*, trans. R. E. W. Madison (1961; Ithaca, N.Y.: Cornell Univ. Press, 1973); J. V. Field, *Kepler's Geometrical Cosmology* (Chicago: Univ. of Chicago Press, 1988); Job Kozhamthadam, *The Discovery of Kepler's Laws: The Interaction of Science, Philosophy, and Religion* (Notre Dame, Ind.: Univ. of Notre Dame Press, 1994).

For several years we have been engaged in a contextual study of Kepler's unification of physics and astronomy. In the course of this project, we have become persuaded that theology plays a central role in Kepler's scientific thinking. Indeed, theology plays several distinct roles in the reception of Copernicus's work. The agenda of Lutheranism indirectly helped to spread the new science, and Kepler was heir to a Lutheran project that succeeded in publicizing Copernican astronomy. But in Kepler's astronomy religious ideas contribute directly to what are now considered scientific achievements: the defense of Copernicanism and the discovery of the laws of planetary motion. In what follows we will briefly review the historical and intellectual background needed to situate Kepler's work in his time; we will then argue that Kepler's first book cannot be understood without acknowledging its religious dimensions and go on to show that similar issues underlie Kepler's demonstration that the orbit of the planet Mars is an ellipse.²

II. COPERNICUS AND THE PHYSICAL BASIS OF ASTRONOMY

When Copernicus's *De Revolutionibus* appeared in 1543, the main dispute in astronomy was between the Averroists, who denied the reality of epicycles and eccentrics based on arguments from Aristotle's physics, and mathematical astronomers, who supported the *theorica* textbook tradition and regarded epicycles and eccentrics as indispensable for predicting the positions of celestial bodies.³ The Averroists insisted that the heavens were divided into a series of concentric orbs, all centered on the earth. Mathematical astronomers followed a construction that came into wide use with Georg Peurbach's *New Theories of the Planets* (*Theoricae novae planetarum*,

On the more specific question of the role of religion in Kepler's thought, see: Jürgen Hübner, *Die Theologie Johannes Keplers zwischen Orthodoxie und Naturwissenschaft* (Tübingen: Mohr, 1975); and Richard S. Westfall, "The Rise and Decline of Orthodox Christianity: A Study of Kepler, Descartes and Newton," in *God and Nature: Historical Essays on the Encounter between Christianity and Science*, ed. David C. Lindberg and Ronald L. Numbers (Berkeley and Los Angeles: Univ. of California Press, 1986), pp. 218–55. In contrast, Robert S. Westman, "The Copernicans and the Churches," in *ibid.*, pp. 76–113, especially pp. 96–8, anticipates the view defended in detail in the present essay.

² Historical study of these issues has benefited from the appearance of several recent books, including Gérard Simon, *Kepler astronome astrologue* (Paris: Gallimard, 1979); Bruce Stephenson, *Kepler's Physical Astronomy* (1987; Princeton: Princeton Univ. Press, 1994), and *The Music of the Heavens: Kepler's Harmonic Astronomy* (Princeton: Princeton Univ. Press, 1994). On the religious background to Kepler's thought, see especially Sachiko Kusakawa, *The Transformation of Natural Philosophy: The Case of Philip Melancthon* (Cambridge: Cambridge Univ. Press, 1995); Charlotte Methuen, *Kepler's Tübingen: Stimulus to a Theological Mathematics* (Aldershot, U.K.: Ashgate, 1998); and the new edition of Max Caspar, *Kepler* (New York: Dover, 1993), with new scholarly apparatus by O. Gingerich and A. Segonds. See also Peter Barker, "The Role of Religion in the Lutheran Response to Copernicus," in *Rethinking the Scientific Revolution*, ed. Margaret J. Osler (Cambridge: Cambridge Univ. Press, 2000), pp. 59–88, and Barker, "Kepler's Epistemology," in *Method and Order in Renaissance Natural Philosophy*, ed. C. Methuen, D. Di Liscia, and E. Kessler (New York: Kluwer, 1997), pp. 355–68. The main works of Kepler referred to are: *Prodromus dissertationem cosmographicarum, continens mysterium cosmographicum* (Tübingen: Gruppenbach, 1596), now usually referred to as *Mysterium Cosmographicum*; and *Astronomia nova AITIO-ΛΟΓΗΤΟΣ, seu physica coelestis* (Heidelberg: G. Voegelinus, 1609), referred to in our text as *The New Astronomy*. The standard translations to which we refer are A. M. Duncan, *Johannes Kepler—Mysterium Cosmographicum: The Secret of the Universe* (Norwalk, Conn.: Abaris, 1981); and William H. Donahue, *Johannes Kepler—New Astronomy* (Cambridge: Cambridge Univ. Press, 1992).

³ Peter Barker, "Copernicus and the Critics of Ptolemy," *J. Hist. Astron.* 30 (1999):343–58, and "Copernicus, the Orbs and the Equant," in *Pierre Duhem: Historian and Philosopher of Science*, ed. R. Ariew and P. Barker, in *Synthese* 83 (1990):317–23.

composed about 1460, first published in 1472).⁴ Peurbach (1423–1461) employed eccentric orbs, some of which carried small spheres performing the function of epicycles. This combination was carried by inner and outer orbs of uneven thickness, so that the system of orbs for each planet had inner and outer surfaces centered on the earth. Planets were embedded in the small orb corresponding to the epicycle and physically transported through the heavens by the combined motions of the complete set of orbs.

In addition to agreeing that the center of the entire system was the earth, the two sides also agreed that the heavens were completely filled by the systems of orbs carrying the planets; the outer surface of the system of orbs for one planet fitted exactly into the inner surface of the system of orbs for the next planet beyond it. Claudius Ptolemy (fl. 150) had introduced the assumption that the systems of orbs fitted inside one another like a perfect set of Russian dolls (or the layers of an onion), now generally called “the nesting hypothesis.” In his *Planetary Hypotheses* Ptolemy used the nesting hypothesis to calculate the absolute distances of planets from the earth and to support his overall pattern for the cosmos by showing that the calculated distance to the sun, based on the models for the moon, Mercury, Venus, and the sun in his *Almagest* (with minor modifications), was independently confirmed by measurements of the solar distance based on parallax. Although the source of these ideas in Ptolemy’s work was unknown in Europe at the time of Nicholas Copernicus (1473–1543), the nesting hypothesis and the corresponding scheme for distances (as well as for the planetary sizes) were well known through Arabic intermediaries and completely assimilated into Western astronomy and cosmology.⁵

The publication of Copernicus’s work did not immediately change prevailing views on the physical basis of astronomy. Mathematical astronomers, including many Lutherans, saw Copernicus as a natural ally in their conflict with Averroist natural philosophers. They adopted those among Copernicus’s innovations that did not challenge their basic understanding of the structure of the cosmos. For example, Ptolemy’s model for the motion of the moon with an epicycle and a “crank mechanism” produced a dramatic variation in the moon’s apparent diameter as it traveled around the earth, contrary to the appearances—a long-standing difficulty that Averroists could point to as evidence against the existence of an epicycle for the moon.⁶ Copernicus’s double-epicycle model for the moon eliminated gross variations in distance and produced an acceptable variation in the moon’s apparent size. So Copernicus’s lunar model could be safely adopted without raising further cosmological questions. In Germany, the Lutheran leader and educator Philipp Melanchthon (1497–1560) included it in a physics text, and in Italy Giovanni Antonio Magini (1555–1617) supplied a *theorica*-style version generated by three-dimensional orbs. The planetary models could also be used for making calculations without endorsing Copernicus’s heliocentric cosmology. Melanchthon’s protégé, Erasmus Reinhold

⁴ E. J. Aiton, “Peurbach’s *Theoricae novae planetarum*: A translation with commentary,” *Osiris* 3 (1987):5–44. A facsimile of the 1472 edition appears in Regiomontanus, *Opera collectanea* (Osna-brück: Zeller, 1972).

⁵ Bernard R. Goldstein, *The Arabic Version of Ptolemy’s Planetary Hypotheses* (Philadelphia: American Philosophical Society, 1967).

⁶ See, e.g., Bernard R. Goldstein, “Remarks on Gemma Frisius’s *De radio astronomico et geometrico*,” in *From Ancient Omens to Statistical Mechanics*, ed. J. L. Berggren and Bernard R. Goldstein (Copenhagen: Univ. Library, 1987), pp. 167–80, especially pp. 172 and 176–7.

(1511–1553), made Copernicus's mathematical models the basis for a new set of astronomical tables, the Prutenic Tables, published in 1551, and played a leading role in spreading Copernicus's fame without endorsing his cosmic scheme.⁷

III. COPERNICANISM AND THE PROBLEM OF METHOD

In the period up to the beginning of Kepler's career, only a few scholars adopted Copernicus's cosmology.⁸ The chief obstacle to general acceptance was posed by the Aristotelian standards of demonstration applied uniformly in all sciences by the sixteenth century, and especially the technique then known as *regressus*. Briefly, a *regressus* consisted of three sets of arguments. The first, also called arguments *a posteriori*, derived the description of an effect from the description of one of its possible causes. In astronomy the "effects" to be explained were the positions of heavenly bodies, and *a posteriori* reasoning led to the construction of hypotheses that "save" these "appearances."⁹ For example, in the case of the sun it was well known that *a posteriori* reasoning led to two possible hypotheses: a concentric circle carrying an epicycle or an eccentric circle.

The second stage in a *regressus* eliminated alternatives, leaving only one, identified as the "true cause" of the original effect. This process was variously described as *consideratio* or *negotatio*. To succeed, it often appealed to information or explanatory principles beyond the original subject matter. Thus Aristotle in effect appealed to principles from geometry to show that only a spherical earth would cast a circular shadow on the moon during a lunar eclipse, and Ioannes Pena (1528–1558) appealed to optics and geometry to show that only a comet which was a physical lens would project a tail of light rays on a great circle away from the sun.

The third stage in a *regressus* was an argument *a priori* that assumed the newly discovered cause and derived the original effect from it by deduction. It is important to remember that the meanings sixteenth- and seventeenth-century authors attached to *a posteriori* and *a priori* had nothing to do with the modern meanings, which date approximately from the work of Immanuel Kant. For authors in our period, *a posteriori* means "reasoning from effects to (possible) causes," while *a priori* means

⁷ Philip Melanchthon, *Initia doctrina physicae*, in *Corpus Reformatorum: Philippi Melanchthonis Opera quae supersunt omnia*, ed. C. G. Bretschneider, 87 vols. (1549; Halle: Schwetschke, 1834–1860), vol. 13, cols. 179–412, especially col. 244 (henceforth CR); Erasmus Reinhold, *Prutenicae tabulae coelestium motuum* (1551; Wittenberg: M. Welack, 1585); Giovanni Antonio Magini, *Novae coelestium orbium theoriae congruentes cum observationibus N. Copernici* (Venice: D. Zenarius, 1589). On the reception of Copernicus, see especially Owen Gingerich, "The Role of Erasmus Reinhold and the Prutenic Tables in the Dissemination of the Copernican Theory," in *Colloquia Copernicana II: Etudes sur l'audience de la théorie héliocentrique*, ed. J. Dobrzycki, *Studia Copernicana* 6 (Wrocław, Poland: Ossolineum, 1973), pp. 43–62; Robert S. Westman, "The Melanchthon Circle, Rhetoric, and the Wittenberg Interpretation of the Copernican Theory," *Isis* 65 (1975):165–93; Peter Barker and Bernard R. Goldstein, "Realism and Instrumentalism in Sixteenth Century Astronomy: A Reappraisal," *Perspect. Sci.* 6 (1999):232–58; and Barker, "Lutheran Response to Copernicus" (cit. n. 2).

⁸ Robert S. Westman, "The Astronomer's Role in the Sixteenth Century: A Preliminary Survey," *Hist. Sci.* 18 (1980):105–47, especially n. 6, p. 136. We would amend this list chiefly by adding R. Gemma Frisius, ("Epistola ad I. Stadius," in I. Stadius, *Ephemerides novae et exactae ab a. 1554 ad a. 1570* [Coloniae Agrippinae: A. Birkmann, 1556], pp. a1r–a2v; see also Goldstein, "Gemma Frisius" [cit. n. 6]) and by noting that Christoph Rothmann abandoned Copernicanism after visiting Tycho Brahe. On Rothmann see Bernard R. Goldstein and Peter Barker, "The Role of Rothmann in the Dissolution of the Celestial Spheres," *Brit. J. Hist. Sci.* 28 (1995):385–403.

⁹ Barker and Goldstein, "Realism and Instrumentalism in Sixteenth Century Astronomy" (cit. n. 7).

“reasoning from the one, true cause to an effect.” Thus, an *a priori* demonstration of the shape of the earth’s shadow began from the assumption that the earth was a sphere, and Pena’s *a priori* demonstration of the antisolarity of comets’ tails began from the assumption that a comet is a spherical lens.¹⁰

In astronomy, it was generally believed that the second and third stages of the *regressus* could not be completed by earthbound observers. This has led to a number of interpretative problems for modern commentators. The frequent *a posteriori* proofs in astronomy have been misread by some commentators as instances of hypothesis testing in the modern sense. The absence of *a priori* arguments (or, in other words, causal proofs) has also led to the common misconception that astronomy was a fictionalist discipline at a time when physics was clearly realist and astronomy was subordinated to physics. What both of these positions ignore is that the whole pattern of *regressus*, with its goal of causal knowledge, remained the ideal in all sciences, including astronomy.¹¹ One major reason why Copernicus’s few supporters in the period leading up to Kepler accepted his cosmic scheme was that they believed it offered causal proofs in astronomy where none had been available before. Reiner Gemma Frisius (1508–1555) and Christoph Rothmann (c. 1555–1597) both repeat such arguments. Their Copernican convictions, then, may have been based on the belief that the Copernican scheme was unique in its ability to explain why superior planets retrogress when in opposition to the sun, why the retrogressions vary in size from planet to planet (with planets closer to the earth making larger retrogressions), and similar phenomena.¹²

If it was true that Copernicus’s scheme gave the only explanation for these effects, then this would be tantamount to showing it was the one true cause of the celestial “phenomena.” Evidently, most European astronomers were unconvinced, and the largest group favoring Copernicus accepted Reinhold’s interpretation, using the mathematics and passing over the cosmology. But the introduction by Tycho Brahe (1546–1601) of a geo-heliocentric cosmic scheme in 1588 removed any simple claim to uniqueness for Copernicus. There were now two possible explanations for all the phenomena claimed as evidence for Copernicus and against Ptolemy. From being a candidate for *a priori* status, Copernicus’s scheme now became just one of two *a posteriori* explanations for the phenomena of planetary positions. To vindicate either position the second and third stages of the *regressus* needed to be completed. Tycho himself used arguments establishing the immobility of the earth to undermine Copernicus. At least one Copernican, the German astronomer Christoph Rothmann, changed sides.¹³ But the majority position in mathematical astronomy remained Ptolemaic.

In 1596, when Kepler began his career with the publication of *The Sacred Mystery of the Cosmos* (*Mysterium Cosmographicum*), he legitimately regarded Copernicus as having achieved no more than *a posteriori* demonstrations. He then set out to provide the *a priori* demonstrations of astronomical phenomena promised by Coper-

¹⁰ On the use of the terms *a priori* and *a posteriori* in this period see Barker and Goldstein, “Realism and Instrumentalism in Sixteenth Century Astronomy” (cit. n. 7); and Barker, “The Lutheran Response to Copernicus” (cit. n. 2).

¹¹ Barker and Goldstein, “Realism and Instrumentalism in Sixteenth Century Astronomy” (cit. n. 7).

¹² Barker, “The Lutheran Response to Copernicus” (cit. n. 2).

¹³ Goldstein and Barker, “The Role of Rothmann” (cit. n. 8), especially pp. 397ff.

nicus but not yet delivered. He would do this not by completing the three steps of the *regressus* but by using intellectual resources provided by his training in Lutheran theology to proceed directly to the *a priori* portion of the demonstration. The pattern of his argument would simultaneously rule out both Ptolemaic and Tychonic alternatives, and his goal was to show that Copernicus's scheme was nothing less than God's plan for the world.

IV. COMETS AND THE SUBSTANCE OF THE HEAVENS

Although Tycho Brahe's system attracted many previous adherents of Ptolemaic astronomy, its introduction required considerable change in the ontology of the heavens—a change that caused the author of the system some difficulty. Using the Copernican distances of planets from the sun, when Mars was closest to the earth it was considerably nearer than the distance from the earth to the sun. Tycho assumed that the sun moved around the earth, while Mars and the other planets moved around the sun. If these motions were represented by systems of orbs, like the orbs used in a *theorica*, then the orbs for Mars and the sun intersected and interpenetrated. This was a physical impossibility on the conventional understanding of the substance of the heavens. In the mid-1580s Tycho abandoned the conventional account, with its system of orbs carrying the planets. Tycho concluded that the substance of the heavens was a continuous fluid of some sort, that the planets moved freely through this medium, and that the orbs of the planets were not physical objects but geometrical constructions representing boundaries in this medium. Observations of two comets played a special role in Tycho's adoption of the new position.

Bright comets appeared in 1577 and 1585. Because of renewed interest in comets earlier in the sixteenth century, the comet of 1577, in particular, was studied by many people. Two observers, Tycho Brahe in Denmark and Michael Maestlin (1550–1631) in Germany, used new techniques to track the distance of the comet from the earth on a daily basis over a period of months. Both concluded that the comet moved in a way which carried it through a series of the geocentric orbs postulated by Aristotle and Ptolemy, but that the motion of the comet was quite consistent with its being carried in an orb centered on the sun and slightly larger than the orb of Venus. For Tycho this motion was ultimately fitted into his new cosmic scheme, with the comet joining the planets in their sun-centered motions, while the sun itself moved around the earth. Maestlin took the more radical step of adopting the Copernican system, although he clearly continued to interpret the ontology of the heavens in the manner familiar from *theorica* and regarded the planets, the comet, and the earth as all being carried by orbs centered on the sun. Maestlin saw additional evidence for his conclusions in the motions of a comet that appeared in 1580.¹⁴

For some years after 1577 Tycho also continued to believe that the planets were carried by orbs. In the next decade, while developing his new cosmic scheme, he puzzled over the intersection of the orbs for Mars and the sun. When a new comet appeared in 1585, it was again subject to intense observation. Shortly afterward,

¹⁴ Michael Maestlin, *Observatio et demonstratio cometae aetherei, qui in anno 1577 et 1578 constitutus in sphaera Veneris, apparuit* (Tübingen: Gruppenbach, 1578); and idem, *Consideratio et observatio cometae aetherei astronomica, qui anno MDLXXX . . . apparuit* (Heidelberg: Jakob Müller, 1581); Tycho Brahe, *De mundi aetherei recentioribus phaenomenis* (Uraniborg, 1588). See also Peter Barker, "The Optical Theory of Comets from [Peter] Apian to Kepler," *Physis* 30 (1993):1–25.

Tycho received a book about the 1585 comet written by Christoph Rothmann, court astronomer to Landgrave William IV of Hesse-Kassel.¹⁵ Rothmann argued convincingly that the 1585 comet was celestial, and that its motion was inconsistent with the substance of the heavens as understood in the Aristotelian tradition. He suggested reviving the Stoic doctrine that the substance of the heavens was a special kind of air. Immediately after receiving this book, Tycho adopted the view that the substance of the heavens was a continuous fluid, solving the problem posed by the intersection of the orbs of Mars and the sun. He gave the first public presentation of his new system of the world in 1588.¹⁶

Returning to the comet of 1577 and the work of Maestlin, the latter's solution to the questions of the comet's position and motion also offered a solution to an outstanding liability of the Copernican system. As mentioned earlier, in both the cosmic schemes of Ptolemaic astronomers and their opponents the Averroists, the heavens were completely filled by sets of orbs in perfect contact with one another, excluding any empty space. This had been achieved by a construction in which the maximum distance for one planet was set equal to the minimum distance for the next planet beyond it. The thickness of each orb was calculated from the maximum and minimum distances for each planet based on Ptolemy's eccentric-plus-epicycle construction. Because there was no systematic connection among the models for different planets in Ptolemy's scheme, it was possible to juxtapose the models in an approved order so that they fitted together exactly. In Copernicus's system this was no longer possible: given his models for each planet, the earth-sun distance fixed the distances for all the planets. Even more embarrassing, if the thicknesses of the orbs for each planet were calculated in the same way as the Ptolemaic equivalents, then there were large gaps between different sets of orbs. As the eccentricity of each planet's motion was very much less than the difference in the mean distances between planets, the space occupied by each orb cluster was very much less than the distances between sets of orbs. In his book on the comet of 1577, Maestlin, in effect, suggested that there was a natural explanation for the gaps in Copernicus's system. Comets are part of the heavens too and require their own orbs to carry them around the sun. The comet of 1577 was carried by an orb system outside the orbs of Venus but inside those carrying the earth-moon system. Perhaps all the gaps were filled by comets.¹⁷

Kepler became Maestlin's student at Tübingen and addressed all the issues we have reviewed so far in the books he wrote later in his life. But confining our attention to astronomy and cosmology eliminates a crucial dimension of Lutheran intellectual life which Kepler would also have acquired at Tübingen, if not before. Why were Lutherans so interested in astronomy that they made publishing the work of Copernicus a special project? Why were Lutherans like Tycho and Maestlin so intensely interested in comets and other celestial phenomena? A large part of the answer is to be found in astrology, which in turn is an instance of the special Lutheran attitude toward the natural world. The great Lutheran teacher and educational reformer Philip Melanchthon had set the pattern for later Lutheran natural philoso-

¹⁵ C. Rothmann, *Descriptio accurata cometae anni 1585* (composed in 1586), in Willebrordus Snellii *descriptio cometae, qui anno 1618 . . .* (Louvain: Elziviriana, 1619).

¹⁶ Brahe, *De mundi aetherei* (cit. n. 14); Goldstein and Barker, "Role of Rothmann" (cit. n. 8); Peter Barker, "Stoic Alternatives to Aristotelian Cosmology: Pena and Rothmann," *Rev. Hist. Sci.*, forthcoming.

¹⁷ Maestlin, *Observatio et demonstratio cometae aetherei . . . 1577 et 1578* (cit. n. 14), pp. 38–9.

phers with his early interest in astrology—perhaps as a way of understanding signs from God that the present world was about to end. Melanchthon not only observed comets and planets himself; he included celestial influences in his definition of physics to provide a rational basis for the study of astrology by Lutherans.¹⁸

Providing signs of the impending end of the world, foreshadowed by the Reformation and the conflict with the Roman Catholic Church, was only one manifestation of God's providential governance of the entire universe. Melanchthon endorsed, and his students elaborated, the doctrine that the entire world was a structure established by God for the benefit of the human race. Adopting the argument from design available in ancient sources such as Cicero's *On the Nature of the Gods* (or *De Natura Deorum*), Melanchthon argued that the orderly and law-like pattern of the natural world showed the work of a benevolent designer. The regular motions of celestial objects were one of the clearest examples of law-like behavior. Thus, the study of astronomy by Lutherans not only had direct practical application in astrology but, like all study of natural philosophy, led both to the recognition that God existed and to an appreciation of his benevolence.

Although Melanchthon's emphasis on providential design gave a new impetus to natural philosophy, early Lutherans never attempted to found their religious beliefs on a purely rational or empirical basis. Luther himself had distinguished between the provinces of law and gospel. While law—for example, the biblical Commandments—might be comprehended rationally, the basis for salvation was the gospel, revealed knowledge. The study of natural philosophy clearly fell on the side of law; while it might enhance piety it was not sufficient by itself for salvation. Melanchthon provided an epistemological foundation for knowledge of moral law by linking it to the doctrine of the natural light, a special faculty of the intellect that gave access to knowledge engraved on the soul.¹⁹ This access was not limited to moral precepts but also explained the special certainty of mathematical knowledge. When Melanchthon used the term "law" to refer to the regular motions of the planets, he suggested that, like other forms of law, once discovered the laws of planetary motion could be recognized to be eternal truths inscribed on the soul by God. Exactly how one removed the layers of error or sin that prevented ordinary people from recognizing such truths was problematic. Education was one method; empirical observation was another. Melanchthon in fact presented the methodology we have reviewed as a pattern for education—the ascent from observation to possible cause, the reasoning from many possible causes to one true cause, and the subsequent *a priori* demonstration of the original phenomenon.²⁰ The same pattern would also allow the recognition of previously unknown causes of natural phenomena, but in an age when the content of knowledge was to a large extent regarded as stable and approaches to improving knowledge focused on reform by recovering classical learning, the application of *regressus* to discovering new knowledge was not a prominent concern.

¹⁸ On Melanchthon's observation of the comet of 1531, see Kusakawa, *Transformation of Natural Philosophy* (cit. n. 2), pp. 125, 170. On his observations of planets, see Philip Melanchthon, *Initia doctrina physicae* (1549), in CR, vol. 13, cols. 268 and 274; Bernard R. Goldstein, "Levi ben Gerson and the Brightness of Mars," *J. Hist. Astron.* 27 (1996):297–300.

¹⁹ On the doctrine of natural light, see Barker, "Kepler's Epistemology" (cit. n. 2).

²⁰ Philip Melanchthon, *Initia doctrina physicae*, in CR, vol. 13, col. 194, quoted and translated in Barker and Goldstein, "Realism and Instrumentalism in Sixteenth Century Astronomy" (cit. n. 7), pp. 244f.; cf. Kusakawa, *Transformation of Natural Philosophy* (cit. n. 2), p. 185.

V. KEPLER'S EDUCATION

Although Kepler is now remembered as an astronomer, his early education was intended to prepare him to enter the ministry of the Lutheran Church in his native Württemberg. The Lutheran initiatives in education had extended from the founding of state-supported public schools and the reform of universities like Wittenberg and Tübingen to the establishment of seminaries or *Stiftsschulen* to train ministers for the new church. Academically talented students were identified in primary school and tracked into work that prepared them for the ministry. The duke of Württemberg provided scholarships to support promising young students: Kepler was one of them. At some time during his university education at Tübingen (1589–1594) Kepler found himself unable to subscribe to the Formula of Concord which all Lutheran clerics were required to endorse. A university career was also ruled out, since entering the ministry was a precondition for appointment.²¹ Kepler was lucky to find an appointment as a teacher of mathematics and other subjects at the Protestant *Stiftsschule* in Graz, a city in Austria. Here he began his publishing career and a successful campaign to attract patronage. His move into mathematics and natural philosophy should not be permitted to obscure the continuing role of Kepler's religious education both in his personal life and in his intellectual work.

All the ideas we have so far reviewed as typical of the sixteenth-century milieu, and especially of Lutheranism, were in active circulation at Tübingen during Kepler's university years. In particular, Andreas Planer (1546–1607), the professor with special responsibility for Aristotle's *Organon* (that is, his logical works) lectured on the *Posterior Analytics*, including both *a posteriori* and *a priori* demonstrations and the conversion of one to the other (and the same methodology was also discussed by Martin Crusius [1526–1607], professor of Greek, and by Maestlin).²² This was hardly surprising. Students everywhere in Europe, regardless of their confessional allegiance, studied these topics. Lutheran universities were distinguished by the influence of Melancthon, and in particular his ideas on natural philosophy and providence. At Tübingen, Melancthon's views were presented by his student Jacob Heerbrand (1521–1600), who taught theology to both Maestlin and Kepler. But professors of astronomy such as Maestlin and his predecessor Philip Apian (1531–1589) also showed the influence of the Lutheran doctrine of providence.

While Kepler was at Tübingen, Heerbrand was professor of theology in the university seminary until 1590 and succeeded Jacob Andreae (1528–1590) as its chancellor from 1590 to 1599. In Heerbrand's writings the Lutheran doctrine that natural philosophy permits access to a divinely created providential ordering of the world is both clearly formulated and extended. For Heerbrand, the natural world is the book of nature to be read in parallel with the book of Scripture in coming to understand God and his works.²³ Heerbrand singles out as elements of the providential design

²¹ Caspar, *Kepler* (cit. n. 1), pp. 48–50, 213, and 258–64; Hübner, *Theologie Johannes Keplers* (cit. n. 1), pp. 45–59 and 108–11; Methuen, *Kepler's Tübingen* (cit. n. 2), pp. 44–6.

²² See Methuen, *Kepler's Tübingen* (cit. n. 2), pp. 183ff.

²³ "I am concentrating [on the materials which form the basis for the *Mysterium*] so that this may be made public as quickly as possible, to the glory of God, who wishes to be known [*agnoscere*] through the Book of Nature," Kepler to Maestlin, 3 Oct. 1595, *Johannes Kepler Gesammelte Werke*, ed. M. Caspar (Munich: Beck, 1937–), vol. 13, p. 40, lines 2–3 (henceforth KGW). On early modern readings of the book of nature, see James J. Bono, *The Word of God and the Languages of Man*:

the same categories of knowledge highlighted by Melanchthon: the order apparent in the moral law and in the realm of numbers. In the tradition of the argument from design, and foreshadowing a comment that Kepler later made in *The Sacred Mystery of the Cosmos*, Heerbrand insists that the order apparent in these realms cannot be accidental or, as he puts it, "fortuitous."²⁴ Like Melanchthon he takes the appearance of such order to be evidence of the existence of a creator. Equally, if the study of natural philosophy is to lead Lutherans to God, this activity presupposes that the providential order of the world is accessible to the human intellect. God's plan for the world is in principle knowable by man.

Heerbrand endorses and extends Melanchthon's views on providence and the study of nature but without attributing these views.²⁵ Kepler also fails to name Melanchthon as a source. There may be two reasons for this. First, these ideas were in some sense the common property of all Lutherans (and shared to a considerable extent by members of other faiths). Second, in the age of the Formula of Concord, Melanchthon's views on the relation of Lutheranism to other confessions (especially Calvinism) became increasingly suspect within the Lutheran community. Although he was revered after his death as the main author of the Augsburg Confession and the great reformer of education in Germany, his rejection by the Lutheran leadership may have made direct citation something of a liability.²⁶

Maestlin became Kepler's most important teacher, and later his benefactor and friend. Maestlin introduced Kepler to Copernicanism, assisted him in finding his first job, and arranged for publication of his first book, *The Sacred Mystery of the Cosmos*. We have already noted Maestlin's work on the comet of 1577, which both pointed out the difficulty posed for Copernicus's system by gaps between his orbs and proposed a possible solution—filling the gaps with comets. Two other elements of Maestlin's work deserve special mention. The first is his explicit discussion of the status of demonstration in astronomy. Although Maestlin and indeed Crusius (the professor of Greek at Tübingen) allow the possibility of *a priori* demonstrations in the mathematical sciences and some related matters, this does not extend to astronomy. For Crusius, in order to qualify as *a priori*, a demonstration must be unique, the only possible demonstration of the phenomenon in question. This can usually be attained only in the mathematical disciplines. Maestlin claims that his demonstration that comets are above the moon (based on parallax observations and geometrical reasoning) is "necessary" (*ex necesse*), which is very much the same thing as calling it *a priori*.²⁷ But these demonstrations are cosmological, not astronomical in the strict sense, and mathematical reasoning plays an unusually large role in them.

Maestlin himself had endorsed Copernicanism in his treatises on the comets of

Interpreting Nature in Early Modern Science and Medicine (Madison: Univ. of Wisconsin Press, 1995).

²⁴ Methuen, *Kepler's Tübingen* (cit. n. 2), p. 137, n. 82; see Kepler, *Mysterium Cosmographicum*, chap. 14; Duncan, *Secret of the Universe*, p. 156 (last paragraph) (both cit. n. 2).

²⁵ Methuen, *Kepler's Tübingen* (cit. n. 2), pp. 136ff.

²⁶ On the changing status of Melanchthon and his ideas at the time of Brahe and Kepler, see Jole Shackelford, "Rosicrucianism, Lutheran Orthodoxy and the Rejection of Paracelsianism in Early Seventeenth Century Denmark," *Bull. Hist. Med.* 70 (1996):181–204.

²⁷ "Ex quo non probabiliter, sed ex necessitate evincitur, Cometam . . . in summo aethere locum sibi quaesivisse." Maestlin, *Consideratio et observatio cometae aetherei astronomica, qui anno MDLXXX . . . apparuit* (cit. n. 14). Quoted in Methuen, *Kepler's Tübingen* (cit. n. 2), p. 179 and n. 61.

1577 and 1580. But in a disputation delivered at Heidelberg in 1582, addressed to an audience that undoubtedly retained the geocentric view of the universe, he gave a completely orthodox statement of the status of astronomy. *A priori* demonstrations are not available to earthbound observers; hence all work in astronomy must proceed *a posteriori*. Maestlin is therefore not among the early Copernicans (like R. Gemma Frisius or Rothmann) who regarded *De Revolutionibus* as offering *a priori* proofs. Maestlin's own conviction is based initially on his observations of the 1577 comet, which could not be made to fit into a Ptolemaic scheme, whereas they could be made to fit into a Copernican one. But everyone who accepts the method of *regressus* also accepts that *a priori* demonstration is the ideal. So for Maestlin and his students, the challenge to supply *a priori* demonstrations in astronomy remains open.

Second, Maestlin's work also embodies the Lutheran conviction that the study of the natural world, especially astronomy, gives knowledge of the Creator's providential plan. But Maestlin adds a significant refinement: he insists that accuracy in astronomy improves one's knowledge of God and providence. This applies both to the description of the comet and to the "most certain laws of the astronomers" (*certissimis Astronomicis legibus*).²⁸ Getting the numbers right matters. Maestlin also serves as the final link in a chain of transmission that connects the mathematical views of Simon Grynaeus to Kepler. Grynaeus (1493–1541) was a friend and collaborator of Melanchthon (with a special interest in astronomy), and he was responsible for the first Greek edition of Ptolemy's *Almagest*. In 1535 Melanchthon wrote a letter to Grynaeus, intended to be used as a preface to a commentary on Peurbach's new *theorica*.²⁹ This letter may well mark the origin of the Lutheran emphasis on astronomy and astrology as sources of the knowledge of God's providence. It was reprinted in several prominent places—for example, at the beginning of both editions of Erasmus Reinhold's *theorica*, in 1542 and 1553.³⁰ Grynaeus himself argued for the legitimacy of mathematically based arguments in establishing the correct interpretation of observational data. His ideas influenced and were transmitted by three highly regarded Tübingen figures during Kepler's time there. The first of these was Martin Crusius, who taught Kepler Greek and later tried to enlist his help with a commentary on Homer.³¹ The second was Philip Apian, who had been professor of astronomy before Maestlin but had lost the position when he refused to subscribe to the Formula of Concord. Apian, a celebrated astronomer in his own right, was still living in Tübingen when Kepler arrived. Third, Maestlin's remarks on the status of mathematics also show Grynaeus's influence.³²

When Kepler arrived at Tübingen in 1589, Maestlin had been professor of astronomy for six years. Although Maestlin confined himself to teaching orthodox Ptole-

²⁸ Maestlin, *Observatio et demonstratio cometae aetherei, . . . 1577 et 1578* (cit. n. 14), quoted in Methuen, *Kepler's Tübingen* (cit. n. 2), pp. 155, 171, 174; for Maestlin's emphasis on exactness in describing the comet, see p. 174, n. 50.

²⁹ Melanchthon to Grynaeus, Jan. 10, 1535, Letter No. 1239, CR, vol. 2, pp. 814–21.

³⁰ Kusukawa, *Transformation of Natural Philosophy* (cit. n. 2), p. 134. The letter reappeared in *Theoricae novae planetarum Georgii Purbachii Germani ab Erasmo Reinholdo Salveldensi . . .* (Wittenberg: Lufft, 1542), and *Theoricae novae planetarum Georgii Purbachii Germani ab Erasmo Reinholdo Salveldensi . . . Recens editae et auctae novis scholiis in Theorica Solis ab ipso autore* (Wittenberg: Lufft, 1553).

³¹ Caspar, *Kepler* (cit. n. 1), pp. 47–8.

³² For evidence of the careful reading by Apian and Crusius of Grynaeus's views that appear in his edition of Euclid's *Elements*, see Methuen, *Kepler's Tübingen* (cit. n. 2), p. 171.

maic astronomy in the basic classes required of all students, he presented Copernican ideas to his advanced students. When Kepler's religious scruples became an obstacle to his original goal of entering the Lutheran ministry, his intensive studies with Maestlin, together with the recommendation of the Tübingen University senate, enabled him to find a position teaching mathematics at a Lutheran school in the Austrian city of Graz. Here Kepler avoided religious partisanship and began a quest for support in the network of patronage that bound together the Holy Roman Empire, and in which the greatest patron was the emperor, Rudolf II. It is against this background that Kepler produced a spectacular piece of intellectual precocity and self-advertisement, in which he claimed to have uncovered, once and for all, the structure of God's providential plan for the cosmos as a whole, and particularly for the arrangement of the planets. Rather than an exercise in astronomy or a defense of Copernicanism as a novel cosmology, Kepler's first book must be read as essentially theological.

VI. THE *MYSTERIUM COSMOGRAPHICUM*

Kepler's first major publication was his *Mysterium Cosmographicum* (1596). The role of religion is not concealed but indicated in the very title of the book, which has not been well translated. "*Mysterium Cosmographicum*" has usually been rendered "secret of the universe."³³ But "secret" is a bland translation of *mysterium*. The term may well mean "mystery" or "secret," but its central meaning in antiquity was "sacred mystery," the secrets taught to initiates when they entered a religious cult. So the title might be better rendered "The Sacred Mystery of the Cosmos." To Kepler and his audience of philologically acute humanists this meaning would have been evident, if not at once, then as soon as the book was opened. The greetings to the reader announce that the book will reveal "What the world is like, that is, God's cause and plan for creating it," among other wonders (*Quid mundus, quae causa Deo, ratioque creandi*).³⁴ This makes the religious aspect of the work unambiguously clear (and indicates to which religion this sacred mystery belongs).

As is well known, Kepler introduces a geometrical construction based on the five regular Platonic solids to defend the Copernican system. The preface to the reader begins,

I propose, reader, to demonstrate in this little book that the most Good and Great Creator, in the creation of this moving world, and the arrangement of the heavens, referred to those five regular solids, well known from Pythagoras and Plato to our own time, and that to their nature he fitted the number of the heavens, their proportions, and the plan (*ratio*) of their motions."³⁵

Twentieth-century historians have usually been happy to endorse Kepler's defense of Copernicus, although his reasoning is often dismissed as mystical. The source of

³³ For example, by Duncan in his English translation, *Johannes Kepler: The Secret of the Universe* (cit. n. 2); also by A. Segonds in his French version, *Le Secret du monde* (Paris: Belles Lettres, 1984).

³⁴ Kepler, *Mysterium Cosmographicum*, fol. A1v; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 48.

³⁵ Kepler, *Mysterium Cosmographicum*, p. 6; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 62. All translations are by P. Barker unless otherwise indicated.

Kepler's alleged mysticism is apparent in this passage; it is the number mysticism associated with Pythagoras and Plato. The equally prominent reference to the Christian deity has usually passed without comment among historians who took it for granted that real science—for example, the Copernican system—had nothing to do with religion.³⁶

Looking at the passage against the background we have already presented, it is conspicuous that all three issues raised by Kepler—the number, proportions, and plan of the celestial motions—are properly questions in cosmology, and not astronomy understood as the science of determining the positions of planets at any given time. Indeed, even in Ptolemaic cosmology the proportions of the heavens were not established from purely astronomical assumptions.³⁷ It is also significant that Kepler claims in the first line that he will offer demonstrations, a point already stated in the greetings to the reader, where he says that he will consider the cause and plan (*ratio*) of the celestial motions. All these points would be read by contemporaries as the claim that Kepler will conform to the standards accepted in *regressus* demonstrations, and that if he claims to know *the* cause of the motions, he will have to establish his results by *a priori* proof. This is exactly what Kepler goes on to do. Religious ideas from the Lutheran tradition play a foundational role in these demonstrations. The naming of the Creator ahead of Pythagoras and Plato is not superficial piety but indicates the real status of religious ideas in Kepler's demonstrations.

In an autobiographical remark that we have no real reason to doubt Kepler tells us that his original insights came to him while teaching one day in July 1595.³⁸ He was drawing a diagram to show the pattern of great conjunctions for the planets Saturn and Jupiter against the background of the zodiac. As he added lines to the diagram, the figure increasingly resembled a series of triangles inscribed within the circle of the zodiac. These triangles defined a second circle in the clear space at the center. Apparently Kepler immediately associated the gap between the two circles—the inscribed circle and the zodiac circle—with the gaps between celestial spheres in Copernicus's system. He saw the possibility of explaining the gaps as the result of boundaries inscribed and circumscribed around geometrical figures.

Kepler experimented with a variety of constructions, even adding new and unknown planets at one point. But, as Kepler soon realized, there was no satisfactory way to define the intervals by means of polygons, particularly in the case of the huge gap between Mars and Jupiter. On the other hand, there were only five regular three-

³⁶ See, e.g., among the works already cited, Dreyer, *History of Planetary Systems* (cit. n. 1), p. 376: "For the order of [the polyhedra in the *Mysterium*] he gives a great many reasons, one, more fantastic than the other. But we must pass over these curious details"; and p. 410: "Many writers have expressed their deep regret that Kepler should have spent so much time on wild speculations and filled his books with all sorts of mystic fantasies"; Caspar, *Kepler* (cit. n. 1), p. 61: "Consciously or unconsciously, Kepler's thoughts were connected with everything which he had heard and read of Pythagoras and Plato . . . and with that which Christian teaching about God and the world and the position of men regarding both had implanted in him. The time had come when these whirling thoughts of Kepler's took on a distinct form . . . [the *Mysterium*]"; and p. 67: "Five manners of approach to the examination of the world enable him to answer [the fundamental questions of the *Mysterium*]: the aesthetic . . . , the teleological . . . , the mystic, by which he is convinced that 'most causes for the things in the world can be derived from God's love for man'"; Koyré, *The Astronomical Revolution* (cit. n. 1), p. 149: "Kepler's mentality seems very strange to us, and the reasoning inspired by it seems fantastic or even harebrained." Similar quotations can be found elsewhere as well.

³⁷ Goldstein and Barker, "Role of Rothmann" (cit. n. 8), pp. 387ff.

³⁸ Kepler, *Mysterium Cosmographicum*, p. 8; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 65.

dimensional figures. Taking the conservative view that the six known planets were all that existed, five regular solids would provide the correct number of intervals. Kepler was delighted to discover that one particular ordering of the solids (moving outward from the sun: octahedron, icosahedron, dodecahedron, pyramid, cube) filled the gaps in close agreement with the distances that followed from Copernicus's system. It should be remembered that for Kepler these regular solids were mathematical objects rather than physical bodies in the heavens. But it is not only the possibility of recovering (something close to) the correct numerical distances that is striking; it is equally significant that there be only one ordering that achieves this. Again there were difficulties: for example, the ratios for the dodecahedron and the icosahedron are the same, as Kepler was explicitly aware.³⁹ Despite this, by a series of arguments he established a unique ordering of the polyhedra. There was one and only one way of arriving at the correct numbers, and this was the mark of an *a priori* demonstration.

In the first chapter of the book Kepler compares what he is doing with the earlier work of Copernicus. Where Copernicus addressed only astronomy, Kepler says he will deal with cosmology; where Copernicus had been able—in the end—to offer only *a posteriori* demonstrations, Kepler will provide, for the first time, an *a priori* demonstration of the Copernican system of the world.⁴⁰

The uniqueness of Kepler's proof would have been its strongest recommendation to a sixteenth-century methodologist. But perhaps the suspicion lingers (especially in the modern mind) that the correspondence between the nested-solids model and Copernicus's numbers might be a mere coincidence, made plausible by Kepler's personal involvement in Platonic number mysticism. Kepler's result is neither personal, coincidental, nor mystical. Melanchthon's disciple, Georg Joachim Rheticus (1514–1574), had already raised the question of why there are six planets and not some other number in his *Narratio Prima*. He had offered as a reason that six is a perfect number (in the mathematical sense that it is equal to the sum of its divisors other than itself), but he had offered no further explanation of why the Creator should have chosen a perfect number, or why the Creator should have chosen *this* perfect number rather than some other. Rheticus's demonstration is clearly not *a priori*. And in the same passage Rheticus acknowledges but does not resolve the problem of the gaps between Copernicus's spheres, saying that there is "no immense interval" between them.⁴¹ By contrast, Kepler's demonstration that there are six planets answers all these questions and is also unique.⁴²

Could the arrangement of the regular solids discovered by Kepler be coincidental or, to use a term we have already introduced, "fortuitous"? The answer depends on seeing that the proof rests on a theological foundation that is not mystical but the overt, common property of Lutherans and many other contemporary Christians. The world has been constructed by a benevolent Creator, according to a discoverable

³⁹ Kepler, *Mysterium Cosmographicum*, p. 27; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 103.

⁴⁰ Kepler, *Mysterium Cosmographicum*, pp. 13 and 23; cf. Duncan, *Secret of the Universe* (cit. n. 2), pp. 77–9, 97–9.

⁴¹ Rheticus, *Narratio prima* (Danzig: Rhode, 1540), Diii r; idem, *Georgii Joachimi Rhetici Narratio prima*, ed. H. Hugonnard-Roche, J.-P. Verdet, M.-P. Lerner et al. (Wrocław: Ossolineum, 1982), p. 60 (Latin text); p. 113 (French translation); E. Rosen, *Three Copernican Treatises*, 2nd ed. (New York: Dover, 1959), p. 147.

⁴² See also the long discussion added by Kepler as n. 7 to the original preface in the 1621 edition of the *Mysterium Cosmographicum*; Duncan, *Secret of the Universe* (cit. n. 2), p. 70.

plan. That the plan turns out to be essentially geometrical is what makes it discoverable. Kepler and many contemporaries believed that knowledge of geometry had been inscribed on the human soul when it was created. Human beings are therefore uniquely well equipped to discover a geometrical plan for the world. Like knowledge of other mathematical truths and of the moral law, such knowledge is accessible by the natural light of reason (a point that Kepler makes about mathematics in several places).⁴³ Knowledge secured in this way is literally guaranteed by God, so no further epistemological guarantees are needed. Why did God use geometrical solids as the basis for the plan of the world? Because he benevolently wished to provide a means for his creatures to come to know his providential design. Why this design and not some other? God might very well have arranged the solids in a different order, but once the order is chosen it yields a unique set of distances, discoverable *a posteriori* from astronomical observation. This confirms the hint that because the number of planets is one more than the number of regular solids, each solid will be used only once. Ultimately, then, the demonstrations both of the number and the spacing of the planets begin from the assumption that there is a providential plan, and that it is knowable by human beings. Kepler makes this aspect of his work explicit in the many references to the Creator deity and his plan for the world at the beginning of *The Sacred Mystery of the Cosmos*, and it is repeated throughout the book. For example, in chapter 4, the author tells us, "I think that from the love of God towards mankind many causes of things in the world may be deduced."⁴⁴ And it is reiterated in the quotation that closes the final chapter:

And now at last with the divine Copernicus it pleases [me] to cry out: Certainly such is the divine handiwork of the Good and Great [God]; and with Pliny: The immense world is sacred.⁴⁵

To sum up: in his own terms, and by the standards of sixteenth-century methodology, Kepler has good reason to believe that he has discovered God's plan for the world. At the same time he has solved the outstanding problem of the gaps between Copernicus's spheres.⁴⁶ He avoids potentially awkward questions about the physical connections between spheres by adopting an air-like continuous fluid as the substance of the heavens. Like Brahe he treats the spheres defined by his cosmic scheme as geometrical boundaries in a continuous physical substance. The abandonment of spheres that physically transport the planets immediately called attention to the question of what moves the planets. This becomes one of the main issues addressed in the *New Astronomy* of 1609. Moreover, Kepler never seriously questions that the cosmos is finite and spherical in shape. In perhaps the best-known theological pas-

⁴³ See, e.g., Kepler, *De Quantitatibus*, cited in Barker, "Kepler's Epistemology" (cit. n. 2), p. 360.

⁴⁴ Kepler, *Mysterium Cosmographicum*, p. 27; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 106.

⁴⁵ Kepler, *Mysterium Cosmographicum*, p. 82; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 223. The passage ends, "and with Pliny: The immense world is sacred, the whole considered as a whole, yea verily itself the whole, finite and resembling the infinite." The sacred whole invoked by Pliny is, of course, the Stoic cosmos.

⁴⁶ Kepler admitted that the "fit" between his theory of the regular solids and the data is not exact but noted "how greatly unequal the numbers would have been, if this undertaking had been contrary to Nature, that is, if God himself at the Creation had not looked to these proportions" (*Mysterium Cosmographicum*, p. 50; Duncan, *Secret of the Universe* [cit. n. 2], p. 157). But Kepler believed that in his *Harmonice Mundi* (1619) he had eliminated the remaining discrepancies.

sage from *The Sacred Mystery of the Cosmos* Kepler claims that the cosmos is literally an image of God:

And the three most important things, of which I persistently sought the causes why they were so and not otherwise, were the number, size and motion of the orbs. That beautiful commensurability (*harmonia*) of static objects: the sun, the fixed stars, and the intervening medium [on the one hand] with God the Father, the Son, and the Holy Spirit [on the other], made me dare this.⁴⁷

Although this structure is secured by similar convictions connecting geometry, the plan for the world, and the nature of the deity, Kepler makes no subsequent claim that it can be demonstrated *a priori*, perhaps because there seems to be no way to establish that the correspondence he proposes is unique.

Like the regular-solids construction, the claim that the cosmos follows the pattern of the Trinity achieves at best an explanation for the static structure of the created world. Kepler is less successful in accounting for the way the planets move. This is perhaps a limitation imposed by using knowledge of geometry as the basis for *a priori* demonstrations. Getting the details of planetary motion right becomes Kepler's major project, occupying him for most of the first decade of the seventeenth century and leading to his *New Astronomy*. In this book he answers the question of the causes of planetary motion and, at the same time, he specifies the pattern of reasoning that led him to them. This book is remembered today for describing Kepler's discovery of the first two laws of planetary motion. As we will see, Kepler makes special use of a second common form of demonstration, to move beyond the static results of *The Sacred Mystery of the Cosmos*, which were achieved in accordance with the *a priori* portion of the *regressus* method. Although the *New Astronomy* is seldom seen as a book with religious content, the Lutheran providential view of nature again underlies the reasoning employed here. It is to this second pattern of reasoning that we now turn.

VII. EXEMPLUM AND THE ARGUMENTS FOR KEPLER'S PHYSICAL PRINCIPLES

The second piece of contextual information needed to understand Kepler's arguments is the meaning of the term *exemplum*. Like *regressus*, the early modern discussion of the *exemplum* pattern of argument is rooted in Aristotle's logic, specifically remarks in the *Prior Analytics* modified by its sixteenth-century adherents. For our purposes, the most significant of these adherents is Melanchthon, whose works on rhetoric and dialectics were enormously influential, especially in Lutheran universities, including that attended by Kepler.

In the *Prior Analytics* Aristotle distinguishes three modes of inference, as follows:

We have an "example" when the major term is proved to belong to the middle by means of a term that resembles the third. It ought to be known both that the middle belongs to the third term and that the first belongs to that which resembles the third. . . . Clearly then to argue by example is neither like reasoning from part to whole, nor like reasoning from whole to part, but rather reasoning from part to part when both particulars are subordinated to the same term and one of them is known. It differs from induction

⁴⁷ Kepler, *Mysterium Cosmographicum*, 1596, p. 6; cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 62.

because induction, starting from all particular cases proves (as we saw) that the major term belongs to the middle and does not apply the syllogistic conclusion to the minor term, whereas argument by example does make this application and does not draw its proof from all the particular cases.⁴⁸

In Greek the term corresponding to “example” in this translation is *paradeigma*, and it is usually translated into Latin as *exemplum*.⁴⁹

In Aristotle *exemplum* may be merely a mode of inference; for Melanchthon and Kepler it is also an indication of the existence of a universal rule or law and, as such, part of God’s providential plan. In his *Erotemata Dialectices* (originally published in 1547, frequently revised and reprinted) Melanchthon repeats Aristotle’s division of inferences into syllogisms, inductions, and *exempla* and goes on to say, “*Exempla* are therefore reminders about some universal rule or law, which connects similar things.”⁵⁰ The examples given to illustrate this type of inference involve moral prohibitions:

The greatest part of the human race perished in the Flood on account of licentiousness [*libidines*]; therefore, without doubt licentiousness will be punished [at the present time].⁵¹

To make the structure of this argument clearer, note that Noah’s Flood was a singular event, brought about, according to Melanchthon, by the licentiousness of the human race. Similar events with similar causes were the destruction of Sodom and of Thebes. It is possible to reason legitimately from these singular cases to another singular case—similar excesses today will also be punished—because there is a moral law that such behavior is wrong, and it is known that God punishes those who transgress the moral law. So a clearer statement might be:

The licentious behavior of humans before the Flood was punished; therefore, licentious behavior today will be punished.

Or, recasting the *exemplum* as a syllogism:

All licentious behavior is punished by God; therefore, licentious behavior today will be punished by God.

Here the first universal premise states the (combination of) moral laws that render the inference between singular instances valid.

The same argument structure recurs frequently in Kepler. A particularly clear and

⁴⁸ Aristotle, *Prior Analytics*, trans. A. J. Jenkinson (Oxford: Clarendon, 1928), II.24, 68b38–69b19.

⁴⁹ E.g., see *Aristoteles Organon seu libri ad Dialecticam attinentes* . . . , trans. F. Caesius (Venice: Apud Hieronymum Scotum, 1552), fols. 103r–103v. In addition to the use of the term *exemplum* by Melanchthon and Kepler (see the following paragraphs), this pattern of argument is widely discussed in sixteenth- and early seventeenth-century texts on logic and dialectics. Two examples are Eustachius à Sancto Paulo, *Summa philosophia quadripartita* (Coloniae: Zetzner, 1629), pt. 1 (Logic), p. 168, who treats it as a fallacy; and Theophraste Bouju, *Corps de toute la philosophie devise en deux parties* (Paris: M. Orry, 1614), p. 73, who accepts it as a nonfallacious pattern of argument.

⁵⁰ *Erotemata Dialectices*, CR, vol. 13, cols. 621–24, col. 622: “Sunt igitur exempla communefactiones [i.e., reminders] de aliqua universali regula seu lege, quae complectitur similia.”

⁵¹ *Ibid.*, col. 622.

brief example follows an argument that the planets vary in receptivity to the solar virtue (on account of which their motions are slower than the rotation of the sun), and that this receptivity increases with their proximity to the sun:

and so in order, all the way to MERCURY, which undoubtedly from the example of the superior [planets], yet again will itself be slower than the power that conveys it. . . . This analogy teaches that there resides in all planets, and also in Mercury itself, the lowest, an inherent material force of extricating itself a little from the orb of the solar virtue.⁵²

Here the *exemplum* argument is:

Saturn, Jupiter, Mars, and Venus move more slowly than the power that conveys them (the solar virtue); therefore Mercury will move more slowly than the power that conveys it.

Kepler, however, is interested in establishing the general rule or law that licenses the *exemplum*: unlike Melanchthon's example, it is not one already known. The syllogistic version of the argument would therefore be:

In all planets there resides a power capable of resisting the effects of the solar virtue (in consequence of which they will move more slowly than the power that conveys them); therefore, in Mercury there resides a power capable of resisting the effects of the solar virtue (in consequence of which it will move more slowly than the power that conveys it).

In both Melanchthon and Kepler, by inspecting a successful *exemplum* argument, we may establish a principle used by God to govern the world.

As these illustrations show, for Melanchthon the primary meaning of *natural law* is a moral law or principle, engraved in the human soul by its Creator and accessible to all through the exercise of that faculty of the soul or mind called the *natural light*. Other principles similarly accessible include the fundamental truths of mathematics. The moral principles were established by God to ensure a stable and harmonious social world for the human race. However, human social life requires a stable physical environment. As already indicated, Lutherans like Melanchthon and Kepler believed that the physical universe had been established in a way and according to a pattern intended for the benefit of mankind. Thus, physical laws, including those to be found in astronomy, were part of the providential plan.⁵³ Melanchthon uses the

⁵² Kepler, *Astronomia Nova* (cit. n. 2), pp. 174–5: “[E]t sic consequenter, usque ad MERCURIAM, qui procul dubio ad exemplum superiorum, etiam ipse tardior erit, virtute quae ipsum vehit. [p. 175] Docet hinc analogia statuere, omnibus PLANETIS, ipse etiam MERCURIO humilimo, inesse vim materialam sese explicandi nonnihil ex orbe virtutis SOLARIS.” Cf. Donahue, *New Astronomy* (cit. n. 2), p. 388.

⁵³ On Melanchthon's concept of natural law and providence, see Kusakawa, *Transformation of Natural Philosophy* (cit. n. 2), pp. 124–73. For additional information on Kepler's knowledge of this tradition, see Methuen, *Kepler's Tübingen* (cit. n. 2). On Kepler's use of these doctrines, see Barker, “Lutheran Response to Copernicus” (cit. n. 2).

term “law” to refer to the pattern of motion of heavenly bodies.⁵⁴ Kepler speaks the same way and also uses the terms “law” and “rule” to refer to the two basic physical principles introduced in the *New Astronomy*: the distance–velocity relation, and the reciprocation rule (called “libration” by Kepler).⁵⁵

Since at least the time when he composed *The Sacred Mystery of the Cosmos*, Kepler relied upon a version of the natural light doctrine to safeguard knowledge of the divine plan.⁵⁶ He later used the term “archetype” to designate the geometrical basis of the plan. Rules or laws such as the distance–velocity relation or the reciprocation rule are not at the same level as these geometrical archetypes. Valid *exemplum* inferences may be taken to establish the existence of a genus to which all its instances (the *exempla*) belong as species, and this may be seen as a law or regularity. The essential difference between archetypes and laws is that the discovery of archetypes depends on mathematical knowledge alone, whereas the discovery of laws requires an investigation and observation of nature. We further suggest that archetypes display the eternal time-invariant features of the divine plan, whereas *exemplum* arguments are used to discover the laws governing the features of the plan that vary in time, such as the positions, distances, and velocities of the planets.⁵⁷

VIII. THE ARGUMENT OF THE NEW ASTRONOMY

A New Astronomy Based on Causes, or Celestial Physics (*Astronomia Nova AITIO-ΛΟΓΗΤΟΣ, sev physica coelestis*) appeared at Prague in 1609. The book begins with a series of chapters in which the systems of Ptolemy, Brahe, and Copernicus are considered as possible models that may account for Tycho’s extremely accurate positional data for Mars. This is an *a posteriori* investigation of astronomy in the sense that prevailed before Kepler—the aim is to recover the phenomena, not to give a causal account of planetary motion understood realistically.⁵⁸ In chapter 16 Kepler introduces a model that uses an equant with a nonbisected eccentricity, which he calls his “vicarious hypothesis,” that is, a hypothesis to be used provisionally

⁵⁴ Kusakawa, *Transformation of Natural Philosophy* (cit. n. 2), p. 140, quoting a passage by Melanchthon published in 1536: “[T]he surest law regulates the heavenly courses and the whole of nature” ([C]ertissima lege cursus coelestes et totam naturam regere) (CR, vol. 3, col. 114).

⁵⁵ Kepler, *Astronomia nova*, p. 276; Donahue, *New Astronomy*, p. 560 (both cit. n. 2); KGW vol. 3, p. 356, lines 14ff., and “leges librationis” line 17. In the *Epitome*, KGW (cit. n. 23), vol. 7, p. 367, line 34, Kepler again applies “leges” to the reciprocation rule.

⁵⁶ Barker, “Kepler’s Epistemology” (cit. n. 2).

⁵⁷ We are not aware of any commentator who has appreciated Kepler’s appeal to *exemplum* arguments, but R. Martens comes very close: “That the mathematical relation holds in both cases is evidence for the precision of the *analogy* and hence for the archetypal nature of the relation, rather than evidence that magnetic poles cause libration”; R. Martens, “Kepler’s Solution to the Problem of a Realist Celestial Mechanics,” *Stud. Hist. Phil. Sci.* 30 (1999):377–94, especially p. 390. What she calls “analogy” is the *exemplum* argument. But, significantly, she has noticed that Kepler has not asserted that the solar virtue is magnetic; rather, both the solar virtue and magnetism belong to the same genus (to use our terminology): see the *New Astronomy*, chaps. 34 and 57, where Kepler presents his *exemplum* arguments. We intend to treat Kepler’s distinction between “archetype” and “physical law” in greater detail in a subsequent publication. For the moment let it suffice to say that the term “archetype” occurs rarely in Kepler’s writings before 1618.

⁵⁸ The modern understanding is that Copernicus at least attempted to give a realistic—that is, a causal—account of planetary motion. But Kepler’s view is that Copernicus’s work was successful only as an *a posteriori* account, leaving to Kepler the task of completing the *a priori* one: *Mysterium Cosmographicum*, chap. 1, Duncan, *Secret of the Universe* (cit. n. 2), p. 76; Martens, “Kepler’s Solution” (cit. n. 57).

until the true hypothesis is discovered. By the end of the second major part of the *New Astronomy* (ending with chap. 21), Kepler has appealed to the now celebrated 8-minute error in longitude to eliminate the models of Brahe, Ptolemy, and Copernicus, leaving only his vicarious hypothesis as a possible account for the angular positions of Mars.⁵⁹ By prevailing standards in physics, however, Kepler cannot offer a mere geometrical model but must identify the true causes of the motions, if he is to make good on his title's promise to offer "an astronomy based on causes." Kepler points out that using an equant model with nonbisected eccentricity gives correct angular positions, while an equant model with bisected eccentricity seems to be needed to recover the correct distances. But a model that embodies the true causes (and permits *a priori* recovery of the observational data) must correctly assign both an angular position and a distance to the planet at any given time. So by the end of the second major part of the overall argument of the *New Astronomy* there is already a clear indication that the vicarious hypothesis itself cannot be the basis for a causal astronomy.⁶⁰

As we have discussed at length in a previous paper, the third part of the *New Astronomy* (concluding with chap. 40) establishes the distance-velocity law for the case of a planet moving on an eccentric circle. This is the first appearance of the result now called Kepler's Second Law, but the modern form of the Second Law employs an ellipse, not a circle. The result of chapter 40 is therefore not Kepler's Second Law but a step on the way to it.⁶¹ In our previous work we were content to show how establishing the distance-velocity law linked mathematical and physical reasoning. However, that law is connected to more general physical principles through *exemplum*-style inferences.

The first important set of *exempla* in the *New Astronomy* occurs in chapter 34, linking light with the motive power in the sun that drives the planets and establishing the physical basis for the distance-velocity law of chapter 40. From this it follows that there exists a genus of which this law is an instance. Physical laws or principles dealing with other aspects of nature may be recognized as legitimate on the grounds that they share the same mathematical structure and are therefore instances of the same genus. Principles that do not may be recognized as spurious and rejected. Kepler needs *exempla* that share more than mathematical similarities. In chapter 36 Kepler says, "I shall propose to the reader the obviously valid *exemplum* of light"⁶² and adds a clear statement that this instance of *exemplum* indicates not an illustration but a pattern of argument, here called "the argument from similar things."⁶³ Kepler explicitly draws an analogy between the cause of the motive power in the sun and the causes of light and of the magnet. Hence, although he does not claim to know all the physical details of this force in the sun, he can claim that such a force exists, as a species of the genus "forces that attenuate with distance." These arguments

⁵⁹ Kepler, *Astronomia Nova*, chap. 19; Donahue, *New Astronomy* (cit. n. 2), p. 286.

⁶⁰ Kepler, *Astronomia Nova*, chap. 19; Donahue, *New Astronomy* (cit. n. 2), p. 286.

⁶¹ Peter Barker and Bernard R. Goldstein, "Distance and Velocity in Kepler's Astronomy," *Ann. Sci.* 51 (1994):59-73.

⁶² "[P]roponam lectori exemplum lucis plane genuinam, cum in SOLIS corpore et ipsa niduletur, indeque comes huic virtuti motrici in totum mundum emicet," Kepler, *Astronomia Nova*, p. 172; cf. Donahue, *New Astronomy* (cit. n. 2), p. 383.

⁶³ Kepler, *Astronomia Nova*, p. 173; cf. Donahue, *New Astronomy* (cit. n. 2), p. 386. In margin: "Exemplum in Luce." In text: "Ut vis argumenti a simili tanto sit evidentior" [literally: "In order that the force of the argument from a similar thing be that much more evident"].

establish the existence of a force that traverses space, diminishes with distance, and moves the planets. This force is assumed in chapter 40, which establishes the distance–velocity law and concludes the third major section of the *New Astronomy*.⁶⁴

The connections Kepler draws between light, the magnet, and the solar force that moves the planets are not merely analogies but evidence for the existence of an underlying physical principle, established by God as part of the providential plan of the world, covering all physical powers that attenuate as they spread out through space.⁶⁵ Because it is deduced from the properties of one such power, the distance–velocity law of chapter 40 may also be recognized as a principle by means of which God directs the providential plan. The results of chapter 40 place Kepler in a position to answer the questions about the motions of the planets left incomplete in *The Sacred Mystery of the Cosmos* and with the same theological certainty as the results of his first book.

The fourth part of the *New Astronomy* (chaps. 41–60) establishes that the path of Mars is an ellipse, a result now called Kepler's First Law when generalized to all planets. This is the most difficult part of the book both mathematically and conceptually, for Kepler offers few guideposts to his reasoning. Much attention has been paid to Kepler's examination of alternative oval curves in the early chapters of part IV, but the key question is how Kepler argues for the correctness of his own solution. Unless he can show that his solution is correct and *unique* he cannot claim to have derived the motion from its *causes*. The elimination of the ovals is an example of standard *a posteriori* reasoning, but the argument for the uniqueness of Kepler's solution again uses *exemplum* arguments to establish the uniqueness of the ellipse, by showing that this curve and only this curve follows from principles that are part of the providential plan of the world. In order to pass from the *a posteriori* portion of a *regressus* to the *a priori* portion, it is common to appeal to principles from higher disciplines, and here Kepler again appeals to principles that originate in theology to establish the *a priori* character of the ellipse. This is especially clear in the case of the last alternative he considers, a puckered oval he calls the *via buccosa*, or "path in the shape of puffed-out cheeks."⁶⁶

The main physical argument recommences in chapter 56 with the reappearance of an epicycle representing reciprocating motion that was first introduced in chapter 39. The goal of that chapter was to indicate the conceptual difficulties with the simple eccentric model, and a reciprocating motion on an epicycle was introduced to illustrate the difficulty. The distance–velocity law established in chapter 40 governs the motion of the planet in longitude but does not adequately determine the distance of the planet from the sun, and a new principle, the reciprocation law, corrects the length of the radius vector so that both the direction and the distance of the

⁶⁴ It is not claimed that the force that moves the planets is identical in all respects to light. In fact, there are important differences: the motive power in the sun attenuates as the distance, whereas light attenuates as the square of distance (as Kepler emphasizes at the beginning of chap. 36). The genus of which these are species is "powers that attenuate with distance." In the new notes to chapter 16 in the second edition of *Mysterium Cosmographicum* Kepler adds another species to the genus, the power that produces heat: Johannes Kepler, *Mysterium Cosmographicum* (Frankfurt: Erasmus Kemper, 1621), p. 61, n. 7. Cf. Duncan, *Secret of the Universe* (cit. n. 2), p. 171.

⁶⁵ Note that Kepler's title for chap. 36 is "By what measure the motive power of the sun is attenuated as it spreads through the world" (*Astronomia Nova*, chap. 36; Donahue, *New Astronomy* [cit. n. 2], p. 394).

⁶⁶ Kepler, *Astronomia Nova* (cit. n. 2), chap. 58.

planet are now specified. But in the same way that Kepler needed to legitimate the physical basis for the distance–velocity rule, he also needs to show that the reciprocation motion is physically legitimate. Although Kepler does not say so explicitly, he clearly realized that the distance–velocity law alone was inadequate for explaining planetary motion. In the early chapters of part IV, he constructed models with various *ad hoc* assumptions that did not work well and were subsequently rejected. The path of Mars was certainly some kind of oval inside the eccentric circle, but the difficulty was in deciding which oval curve was the correct one. The reciprocation rule then reappears, because Kepler saw that it offered the possibility of being justified as a physical principle.

Chapter 57 argues, again using *exemplum* style inference, that the reciprocating motion represents a natural law (and hence is part of the providential plan). The chapter is entitled “On the Physical or Metaphysical Basis of the Libration Motion,” and a note to the title makes Kepler’s goal even clearer: “By what natural principles a planet may be made to reciprocate as if on the diameter of an epicycle.” Kepler is not presenting analogies in an attempt to persuade the reader of the plausibility of reciprocation; he is looking for similar physical systems which can be used to establish that the reciprocation is a species of a wider genus, and hence a law of nature. The first instance he considers is a circular river and a boat directed by an oar; the direction of the boat varies over time, so that it revolves in twice the periodic time of the planet (twice the time it takes to go once around the river). However, this example is physically unacceptable to Kepler because the faces of the planets should appear to change, while the face of the moon, although it participates with the planets in the motion under discussion, does not change over the course of a month; and, more importantly, the “species” of the sun is immaterial, while the river, oar, and boat are material.⁶⁷ As in the earlier series of *exempla*, Kepler proceeds from the material to the immaterial.

The critical step in the argument is signaled by the marginal note “Exemplum Magneticum.” Kepler argues that a magnetic solar force acting on planets that are magnets will bring about the reciprocation motion.⁶⁸ Previously, in part III, Kepler appealed to one property of a magnet, namely, that its force diminishes with distance, whereas here he appeals to another property of a magnet, namely, that it both attracts and repels. Two points deserve special emphasis: first, Kepler concludes, not that the solar force is magnetic, but that it is a species of the same genus as magnetic force. Kepler notes explicitly that the reciprocation motion obeys the same law as the balance beam or scales.⁶⁹ Second, on the grounds that the physical influence responsible for reciprocation in the motion of a planet is a species of an established

⁶⁷ Kepler, *Astronomia Nova*, pp. 269–70; cf. Donahue, *New Astronomy* (cit. n. 2), 549–50, corresponding to the passages between the marginal notes: “*Exempla naturalia librationum huiusmodi*” and “*Exempli defectus*.”

⁶⁸ Kepler, *Astronomia Nova*, pp. 271–74; cf. Donahue, *New Astronomy* (cit. n. 2), pp. 550ff. In margin: “*Exemplum Telluris*” (p. 271) and “*Exemplum Magneticum*” (p. 272).

⁶⁹ Kepler, *Astronomia Nova* (cit. n. 2), p. 273, with a marginal note by Kepler: “Reciprocation works according to the law of the balance; hence the name ‘Libration.’” In chapter 33 of the *Astronomia Nova*, Kepler introduced the balance beam as a preliminary analogy for the motion of the planets, here invoking terminology drawn from medieval physics: “intension and remission of motion.” Donahue, *New Astronomy* (cit. n. 2), pp. 376, 378; see also John E. Murdoch and Edith D. Sylla, “The Science of Motion,” in *Science in the Middle Ages*, ed. David C. Lindberg (Chicago: Univ. Chicago Press, 1978), pp. 206–64, especially pp. 237ff. In chap. 57 (Donahue, *New Astronomy* [cit.

genus, this influence, whatever it is, can be recognized as part of God's governance of his creation and hence a law of nature. It is this status that shows the reciprocation rule to be the only possible explanation for the corresponding motion of the planet and entitles Kepler to claim that he has found a causal account. This is especially clear in the case of the last alternative to the ellipse, eliminated by Kepler in chapter 58, the *via buccosa*. Here the alternative curve is eliminated, not because it fails to fit the observations but because the ellipse—and only the ellipse—follows from the combination of the distance–velocity law and the reciprocation law. And what makes that a good basis for selecting between otherwise equally successful curves is that the two laws invoked here have already been shown to be parts of the providential plan, by means of *exemplum* inferences. Hence the two real laws presented in the *New Astronomy* are not Kepler's First and Second Laws, as we know them today, but the distance–velocity law and the reciprocation law.⁷⁰

At the beginning of chapter 58, Kepler says, "Throughout this entire work, my aim has been to find a physical hypothesis that not only will produce distances in agreement with those observed, but also, and at the same time, sound equations [i.e., proper corrections to the planet's angular positions], which hitherto we have been driven to borrow from the vicarious hypothesis of chapter 16."⁷¹ The path of the planet, in the sense of its two-dimensional track in both distance and direction from the sun, will be specified by means of the distance–velocity law acting together with the reciprocation rule. This turns out to be the ellipse, which is not therefore a law itself but a consequence of the application of two separate and independent laws. It is often said that Kepler depended on curve fitting, and that because a whole family of curves is observationally indistinguishable from the ellipse, Kepler's argument is not sound.⁷² In fact, Kepler has concluded with a *regressus* argument: he considers

n. 2], p. 566), Kepler cites (pseudo-)Aristotle's *Mechanics* in connection with the law of the lever; for a discussion see Joseph E. Brown, "The Science of Weights," in Lindberg, *Science in the Middle Ages* [cit. n. 69], pp. 179–205.

⁷⁰ The reciprocation law is also called the "versine rule"; Kepler calls it "libration." The "Area Law" for the circle is introduced in chapter 40 as an approximation to the distance–velocity law, but Kepler never gives it the status of a "law." In chapter 59, where Kepler is deriving the ellipse from his two laws, he still appeals to the distance–velocity law (Donahue, *New Astronomy* [cit. n. 2], p. 585, n. 16). The correct relationship between the distance–velocity law and the Area Law was not established by Kepler until the *Epitome Astronomiae Copernicanae*, or *Epitome of Copernican Astronomy* (Linz: Tampachius, 1618–1621), where he indicates that there are two components of motion that lead to the ellipse: one is perpendicular to the radius vector from the sun to the planet, and the other is a reciprocation along the radius vector from the sun to the planet. This modifies his previous explanation in the *New Astronomy* and is equivalent to the Area Law: "Therefore in order to form [the elliptical orbit] two elements of movement are mingled together, as has been demonstrated already: one element comes from the revolution around the sun by reason of one solar virtue; the other comes from the libration towards the sun by reason of another solar virtue distinct from the first." Kepler, *Epitome of Copernican Astronomy*, KGW, vol. 7, p. 377. Cf. E. J. Aiton, "Infinitesimals and the Area Law," in *Internationales Kepler-Symposium, Weil der Stadt 1971*, ed. F. Krafft, K. Meyer, and B. Sticker (Hildesheim: Gerstenberg, 1973), pp. 285–305, especially pp. 303ff.; and Stephenson, *Kepler's Physical Astronomy* (cit. n. 2), pp. 163–65.

⁷¹ Donahue, *New Astronomy* (cit. n. 2), p. 573.

⁷² In a letter to Edmond Halley (1656?–1743) in 1686 Newton wrote that "Kepler knew the Orb to be not circular but oval and guest it to be Elliptical." Quoted in Curtis Wilson, "The Newtonian Achievement in Astronomy," in *The General History of Astronomy: Planetary Astronomy from the Renaissance to the Rise of Astrophysics*, ed. R. Taton and C. Wilson, vol. 2A: *Tycho Brahe to Newton* (Cambridge Univ. Press, 1989), pp. 233–74, especially p. 238; see also Aiton, "Infinitesimals and

two models that account for the data (the ellipse and the *via buccosa*), but only the ellipse can be derived *a priori* by geometrical demonstration and as a result of the combined effect of his two laws (as Kepler demonstrates in chapters 59 and 60).⁷³ To be sure, in chapter 58 Kepler also claims that the ellipse fits the observational data slightly better than the *via buccosa*, but that is not the essential part of the argument, as it shows the ellipse to be only a possible cause of the phenomena (observed positions). Similarly, Kepler indicates in chapter 58 that the ellipse has symmetrical properties lacking in the *via buccosa*, but that too is insufficient to establish that the ellipse is the only possible cause of the phenomena, which is the result Kepler needs. To recapitulate: Kepler can legitimately claim to have offered a causal astronomy, by prevailing sixteenth-century standards, because (1) the ellipse follows uniquely from the distance–velocity law and the reciprocation law; (2) the distance–velocity law and the reciprocation law are individually defensible by *exemplum* arguments; and (3) principles capable of defense in this way are true laws, that is, they are part of the providential plan. In place of the *negotatio*, or elimination of possible causes to identify the one true cause in a conventional *regressus*, Kepler invokes the special status of *exemplum* arguments in Melanchthon and Lutheran natural philosophy to establish that his physical principles are the correct ones, and the one true cause of planetary motion.

Lutheran theology connects the physical arguments and the mathematical arguments. It was the failure to recognize these connections that made it difficult for previous commentators to appreciate the significance of Kepler's physical reasoning and to see the full force of his claim to have achieved a causal astronomy. Modern readers locate two laws in the *New Astronomy*: the so-called Area Law (or Second Law) and the First Law, which defines planetary orbits as ellipses with the sun at one focus. We have seen that Kepler offers two "laws" or "rules" in the course of his book. These are the distance–velocity law produced in chapter 40, and the reciprocation rule or versine rule, especially in chapter 57. The reappearance of the distance–velocity rule in chapter 59 and its restatement as applicable in the case of an elliptical path is not the statement of the "correct" Area Law but rather an integral element in Kepler's final argument for his claim to have given a causal account of planetary motion. "And unless the physical causes that I had taken in the place of the principles had been good ones, they would never have been able to withstand an investigation of such exactitude."⁷⁴ The two laws (distance–velocity and reciprocation) are necessary (in the logical sense) for the causally based account of planetary motion promised in the title: *A New Astronomy Based on Causes*; these causes in fact yield both the distance and the direction of the planet.

the Area Law" (cit. n. 70), p. 300, n. 63; D. T. Whiteside, "Newton's Early Thoughts on Planetary Motion," *Brit. J. Hist. Sci.* 2 (1964):117–37, especially p. 129, n. 42; and idem, "Keplerian Planetary Eggs, Laid and Unlaid, 1600–1605," *J. Hist. Astron.* 5 (1974):1–21, especially p. 14 and n. 41.

⁷³ For different interpretations, see E. J. Aiton, "Johannes Kepler and the Astronomy without Hypotheses," *Jap. Stud. Hist. Sci.* 14 (1975):49–71, especially p. 65; and C. Wilson, "Kepler's Derivation of the Elliptical Path," *Isis* 59 (1968):5–25, especially pp. 17ff.

⁷⁴ Kepler, *Astronomia Nova* (cit. n. 2), chap. 59, p. 295; Donahue, *New Astronomy* (cit. n. 2), p. 591. In the immediately preceding passage, Kepler says, "[I]t [the direction from the sun to the planet] agrees exactly with the vicarious hypothesis, that is, with the observations. And when the fact was established, I was afterwards driven, once I had settled on the principles, to seek the cause of the matter which I have revealed to the reader in this chapter as skilfully and lucidly as possible."

IX. CONCLUSION

In the Lutheran response to Copernicus we see religion playing two important roles. First, the spread of Lutheranism and its educational reforms becomes a vehicle for spreading Copernicus's ideas. A Lutheran (Rheticus, a disciple of Melanchthon) persuaded Copernicus to publish his *magnum opus* and arranged for the production of the book, and other Lutherans used it in teaching astronomy at their universities. But Lutherans regarded Copernicus as a reformer, solving the long-standing problem of nonuniform circular motion in Ptolemaic astronomy, rather than as the discoverer of the true nature of the cosmos. Melanchthon's protégé Erasmus Reinhold and his successors retained Ptolemy's cosmic scheme, even if they adopted Copernicus's mathematical models for calculating the positions of the planets. Although anyone reading *De Revolutionibus* for its mathematical techniques would also have been exposed to Copernicus's heliocentric scheme, Lutherans, in common with the overwhelming majority of other sixteenth-century readers, proved strikingly resistant to Copernicus's cosmology. At worst, it could be argued, Copernicus's book was no more than flotsam carried by the spreading tide of Lutheranism. The vast majority of Lutherans who allude to Copernicus did not find any support in their theology for the new cosmology but only called attention to a set of objections—biblical passages that were understood to exclude the motion of the earth—that partially explains the negative response to Copernicus's cosmic scheme.⁷⁵

Kepler's work, however, shows a second and far stronger connection between religion and science. The books in which this influence appears are not minor or peripheral—they are the book in which Kepler himself stated that he had presented the principal features of his lifetime research program (*The Sacred Mystery of the Cosmos*), and the book in which he claimed to have finally given a true causal account of planetary motion (*A New Astronomy Based on Causes*). Today these books are regarded, respectively, as the first major defense of heliocentrism after the death of Copernicus, and the first statement of the true laws of planetary motion. Both books contain discussions of the causes of planetary motion that are acknowledged ancestors of Newton's theory of universal gravitation. Showing that religion played a role in the reasoning of these books places it at the center of the most important developments in early modern science.

We have suggested that Kepler's causal reasoning cannot be understood except through a prior understanding of two things: his use of *regressus* and his use of *exemplum* reasoning. In both cases his religious convictions inform his use of these patterns of argument and enable him—as he sees it—to achieve results that were inaccessible to his predecessors. The conviction that God has created the world according to an intelligible plan that he, Kepler, has discovered, underlies the claims to knowledge in both *The Sacred Mystery of the Cosmos* and *A New Astronomy Based on Causes*. In the first, it is the confidence that God's geometrical plan for the world is accessible through the natural light of reason that underlies the *a priori* demonstration of the structure of the world and the defense of Copernicus's cosmic

⁷⁵ In Kepler's introduction to the *New Astronomy*, there is a section that begins, "There are, however, many more people who are moved by piety to withhold consent from Copernicus, fearing that falsehood might be charged against the Holy Spirit speaking in the scriptures if we say that the earth is moved and the sun stands still" (Donahue, *New Astronomy* [cit. n. 2], p. 59). Kepler goes on to argue that such fears are baseless.

scheme. In the second, it is the related conviction that *exemplum* arguments reveal the laws by which God governs the providentially ordered world that vindicates the laws of planetary motion.

In *A New Astronomy Based on Causes* Kepler regards what we have called the distance-velocity rule and the reciprocation rule as the true laws, from which the ellipse and the Area Law follow as necessary consequences. These latter are distinguished from other possible patterns, such as the *via buccosa*, and guaranteed as the only possible pattern of planetary motion, because they follow from rules or laws that are known to be part of the providential plan. According to the accepted standards of *regressus* it is Kepler's demonstration that his analysis yields a unique answer that shows it is also sufficient. Kepler can then conclude that he has discovered the one true cause of planetary motion, satisfying the most stringent methodological requirements of his contemporaries and justifying the title of his book. At the same time he completes the Copernican agenda of providing a physically real, that is, causally based, astronomy. It would also have been apparent to his contemporaries that Kepler has scrupulously observed the accepted order of subordination or subalternation in the sciences. His fundamental principles are theological; they are used to guarantee conclusions in physics; and these, in turn, are used to demonstrate results in astronomy.

Kepler is usually credited with discovering three of the earliest scientific laws of the modern period. If we are right, a more historically defensible claim would be that Kepler believed he had discovered the part of God's providential plan that embodied the pattern of the cosmos, and the divine laws by which God regulated its moving parts. The idea of a providential plan, and especially the divine laws that regulate its parts, may therefore be seen as an essential step preceding and preparing the way for the secular concept of a law of nature.